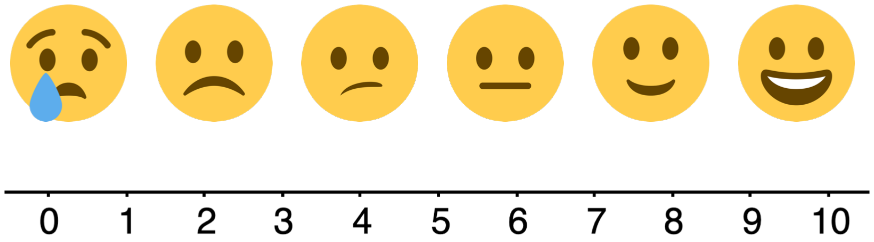
6.10.8 Wrap-Up: Reflection

1. Use the emoji scale to indicate your level of confidence with understanding and using the rules for multiplying and dividing integers.



Unit 6, Lesson 11: Multiplying and Dividing Integers

Benchmark

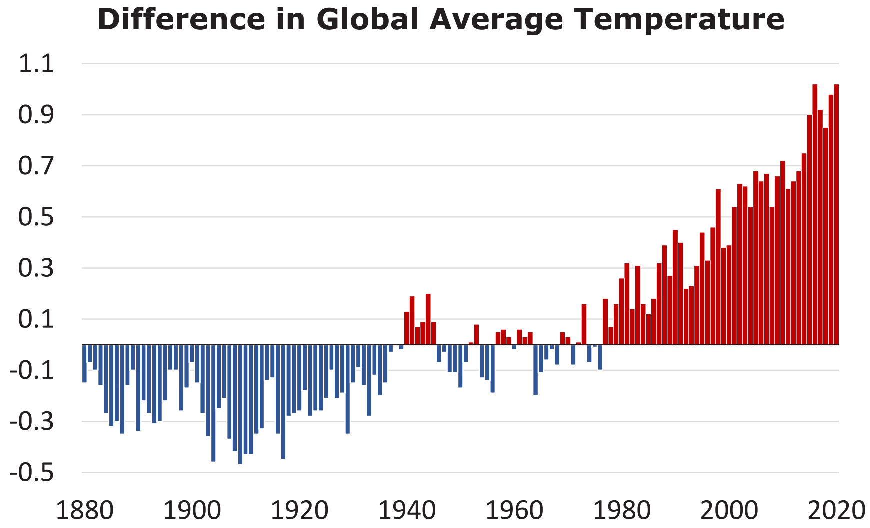
MA.6.NSO.4.2 Apply and extend previous understandings of operations with whole numbers to multiply and divide integers with procedural fluency.

Learning Target

- ☐ I can multiply and divide integers with procedural fluency.

6.11.1 Warm-Up: What's Going On in This Graph?

The graph shows the average global temperatures between 1880 and 2020 compared to the middle of the 20th century. All temperatures are in degrees Celsius (℃).



1. What are you curious about after examining the graph?

2. Imagine this graph is included in a newspaper article. Write a catchy headline that captures what's going on in the graph.

6.11.2 Refresher: Multiplying and Dividing Integers

1. Work with your partner to complete the tables. The last row of each table is missing both the rule and an example equation.

Visual Model of Multiplication	Sign of Product	Example Equation
+ × −	−	
+ × +	+	
		$-14 \cdot 4 = -56$

Visual Model of Division	Sign of Quotient	Example Equation
$\frac{-}{-}$	+	
+ ÷ −	−	
		$\frac{24}{4} = 6$

6.11.3 Collaboration: Multiplying and Dividing Integers

1. The table shows two different products. Work with your partner to find five different factor pairs that, when multiplied, will result in the product. The factors cannot include 1 or -1.

Product	Factor	Factor
(-90)		
42		

2. Explain a strategy or an approach that each partner suggested.

3. Explain which of the products, (-90) or 42, was the easiest to find factor pairs for and why.

4. The table shows two different quotients. Work with your partner to find four different pairs of dividends and divisors that result in the given quotient when they are divided. The numbers 1 or -1 cannot be used.

$$\textit{dividend} \div \textit{divisor} = \textit{quotient}$$

Dividend	Divisor	Quotient
		(-8)
		13

5. Explain a strategy or an approach that each partner suggested.

6. Explain which of the quotients, (-8) or 13, was the easiest to find dividends and divisors for and why.

6.11.4 Guided Practice: Finding Missing Factors, Products, or Quotients

1. Fill in the missing values to make each equation true.

   a. $(-7) \cdot \_\_\_\_\_ = -14$

   b. $\_\_\_\_\_ \div (-5) = 7$

   c. $\frac{-40}{8} = \_\_\_\_\_$

   d. $\_\_\_\_\_ \times (-9) = 63$

2. Consider the equation shown. Phillip said -12 is the missing factor while Alicia claimed it is 12.

$$(-1) \cdot (-2) \cdot \underline{\ ?\ } = 24$$

   Explain which student is correct using rules for multiplying integers.

6.11.5 Your Turn!

1. Find each product or quotient.

-11×9	$-4 \cdot (-12)$	$-100 \div 25$	$\frac{-60}{-5}$
$96 \div (-6)$	$(-5) \times (-13)$	$\frac{84}{-12}$	$(-7)(-7)$
$110 \div (-11)$	$(-6)(-24)$	$9 \times (-15)$	-18×96
$26 \div (-13)$	$150 \div (-3)$	$-39 \times (-4)$	$\frac{-88}{-4}$